
Liar, Liar: Beyond Vocabulary Suppression

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github.com/aryan-cs/liar-liar

Activation-steering vectors for honesty raise truthfulness benchmarks, and this is widely read as evidence that the intervention moves an upstream concept. We ask whether the effect instead rides the model’s output readout — a bias toward honest-coded words rather than a change in what the model computes. We separate the two by projecting a steering vector onto the orthogonal complement of the effective-unembedding rows (the unembedding composed with the RMSNorm Jacobian) for a chosen set of deception-coded tokens. The projected vector has certified zero direct logit effect on those tokens, so any behavior it preserves must travel the indirect path through downstream layers; the preserved fraction ρ measures depth. We show that the statistic is only meaningful once two pitfalls are controlled: a steering magnitude must be selected under a coherence gate, because an over-large coefficient inflates teacher-forced multiple-choice scores while destroying free generation and turning ρ into a ratio of noise; and the unembedding subspace must beat a random subspace of equal rank, or its removal proves nothing. On Llama-3-8B-Instruct we decompose two honesty-steering constructions head to head and find **[empirical verdict: filled from results]**. Code, token sets, certificates, and a formal proof of the construction are released.

1 Introduction

Activation steering has become a standard lever for controlling language-model behavior: add a contrastively constructed vector to the residual stream at a middle layer, and benchmark measures of honesty, refusal, or sentiment shift in the intended direction (Zou et al., 2023; Rimskey et al., 2024; Li et al., 2023; Arditì et al., 2024). The appeal is mechanistic — the intervention appears to reach in and adjust a high-level concept the model uses. For honesty in particular, this reading underwrites a safety story: if a steering vector moves an internal representation of truthfulness, the same vector should generalize across phrasings, domains, and languages.

A deflationary alternative is rarely ruled out. A honesty vector that merely tilts the output distribution toward words like *true* and *honest* and away from *lie* and *deceive* would post the same benchmark gain while changing nothing upstream. The two accounts — an upstream concept versus a readout-level vocabulary bias — predict the same TruthfulQA score and very different out-of-distribution behavior. The standard evaluation does not separate them, because adding a vector at layer ℓ^* moves both the residual the unembedding reads directly and the residual that downstream layers transform first.

We separate the two paths by construction. The first-order logit response to an injected vector v splits into a direct term $\widetilde{W}_U^* v$, the part the readout reads unchanged, and an indirect term $\widetilde{W}_U^* M v$, the part that travels through downstream attention and MLP blocks (Section 3). For a set T of deception-coded tokens, we project the steering vector onto the orthogonal complement of the corresponding

rows of the *effective* unembedding — the unembedding composed with the RMSNorm Jacobian at the readout point. The projected vector $v^\perp$ has exactly zero direct effect on every token in T , certified to machine precision, so any behavior it preserves must travel the indirect path. The fraction preserved, $\rho = \Delta(v^\perp)/\Delta(v_{\text{dec}})$, is a depth statistic: near 0 if the effect was direct readout, near 1 if it was downstream computation.

Two design choices make the statistic trustworthy, and both turned out to matter. First, the ratio is only meaningful when its denominator — the steering effect itself — is real. We select each vector’s operating point under a coherence gate that rejects magnitudes which inflate held-out perplexity, because a large steering coefficient can raise teacher-forced multiple-choice scores while reducing free generation to noise; a sweep that ignores this picks a setting where ρ is a ratio of two noise terms. Second, the unembedding subspace must be shown to be special: we compare projecting out the aligned T against projecting out a random subspace of equal rank. If a random projection erases the effect just as well, the effect was never concentrated in the readout, and a high ρ means nothing.

Contributions. (i) A token-conditional projection that zeroes a steering vector’s direct logit contribution on a chosen token set, with the RMSNorm correction that prior unembedding-orthogonal constructions omit, and machine-precision zero-direct-effect certificates. (ii) A depth statistic ρ with a coherence-gated operating-point protocol and a random-projection control that together make it interpretable, on Llama-3-8B-Instruct. (iii) A head-to-head decomposition of two honesty-steering constructions — contrastive activation addition and the mass-mean direction — showing **[empirical verdict: filled from results]**. All code, token lists, certificates, and the formal proof are released.

2 Related work

Activation steering. Representation engineering adds contrastively constructed vectors to the residual stream to modulate high-level behavior (Zou et al., 2023); contrastive activation addition (Rimsky et al., 2024) and inference-time intervention (Li et al., 2023) are the canonical honesty-adjacent instantiations, and Arditì et al. (2024) show refusal is mediated by a single direction that can be removed from every writing matrix. Tan et al. (2024) document that steering vectors often generalize poorly out of distribution; our depth statistic offers a mechanism-level account of when that should happen.

Concept erasure. LEACE (Belrose et al., 2023) gives the minimum-norm affine map that removes linear decodability of a concept, following iterative nullspace projection (Ravfogel et al., 2020) and rank-constrained adversarial erasure (Ravfogel et al., 2022). Our projection is structurally the LEACE construction with two changes: the erased subspace is read directly off the model’s effective unembedding rather than learned from labels, and the constraint is zero direct logit contribution on a token set rather than zero probe accuracy.

Unembedding geometry. Park et al. (2024) formalize the duality between context-side and unembedding-side concept directions under a causal inner product; our rank-one variant projects against the unembedding-side honesty direction, and our Mahalanobis-weighted variant recovers their inner product. Marks and Tegmark (2024) construct truth directions by mass-mean differences, which we use both as an alternative steering vector and as the readout-side direction of the rank-one test. The direct logit-attribution path through the residual stream is the lens of Elhage et al. (2021) and nostalgebraist (2020).

Closest precedents. Three works overlap most directly. Venkatesh and Kurupath (2026) prove steering vectors are non-identifiable up to perturbations in the null space of the full activation-to-logit Jacobian; our subspace is the readout-restricted version, and where they establish ambiguity we measure what the ambiguity carries. Nadaf (2026) show function vectors steer behavior the logit lens cannot decode, establishing the off-readout channel in the in-context-learning setting. The

unembedding-steering benchmark of [van Deventer \(2024\)](#) implements W_U -orthogonal steering for sentiment on Gemma-2-9B. None applies the construction to honesty steering, none corrects for the RMSNorm Jacobian, and none reports a quantitative depth decomposition with certificates; those are the contributions here.

3 Method

3.1 Setup and notation

Let $\mathcal{M}$ be a decoder-only transformer with L layers, residual width d , and vocabulary size V . The residual stream $h_i^{(\ell)} \in \mathbb{R}^d$ is updated additively by each block, and the logits at the final position are

$$\ell(h_n^{(L)}) = W_U \cdot \text{RMSNorm}_\gamma(h_n^{(L)}), \quad \text{RMSNorm}_\gamma(z) = \gamma \odot \frac{z}{\sqrt{d^{-1}\|z\|_2^2 + \varepsilon}}, \quad (1)$$

with $W_U \in \mathbb{R}^{V \times d}$ the unembedding and $\gamma \in \mathbb{R}^d$ a learned gain ([Zhang and Sennrich, 2019](#)). A steering intervention adds a vector αv_{dec} to the residual stream at a middle layer ℓ^* at every position. We use the contrastive mean-difference construction ([Rimsky et al., 2024](#); [Zou et al., 2023](#)): given prompts rendered under an honest and a deceptive system prompt, v_{dec} is the difference of mean final-token residuals at ℓ^* .

Effective unembedding. For a perturbation δz at the readout point z , the first-order logit response is governed not by W_U but by the composition with the RMSNorm Jacobian,

$$\widetilde{W}_U(z) := W_U J_\gamma(z), \quad J_\gamma(z) = \frac{1}{\sigma(z)} \text{diag}(\gamma) \left(I_d - \frac{zz^\top}{d\sigma(z)^2} \right), \quad (2)$$

where $\sigma(z) = \sqrt{d^{-1}\|z\|_2^2 + \varepsilon}$. Projecting against raw W_U rows, as in prior unembedding-orthogonal constructions, leaves a residual direct effect through the gain and the rank-one term. We average J_γ over a calibration set of readout points and write $\widetilde{W}_U^*$ for the resulting matrix.

3.2 Token-conditional orthogonalization

A global version of the test — a perturbation invisible to *every* token — is impossible: for every model we consider, $V > d$ and W_U has full column rank, so $\ker(W_U) = \{0\}$ (Proposition 3.1 of the supplementary proof). The construction must restrict the readout to a chosen token set.

Fix a deception-coded token set $T \subset \{1, \dots, V\}$ with $k = |T| \ll d$, and let $A = \widetilde{W}_U^*[T, :] \in \mathbb{R}^{k \times d}$ be the corresponding rows. With A^+ the Moore–Penrose pseudoinverse,

$$P_T^\perp = I_d - A^+A, \quad v^\perp = P_T^\perp v_{\text{dec}}, \quad v^\parallel = v_{\text{dec}} - v^\perp. \quad (3)$$

By construction $A v^\perp = 0$: the projected vector has exactly zero first-order logit contribution at every token in T . Among all vectors satisfying that constraint, $v^\perp$ is the one closest to v_{dec} in Euclidean norm (the LEACE-style minimum-norm characterization; Theorem 8.1 of the supplementary proof, after [Belrose et al., 2023](#)).

3.3 Direct and indirect paths

Writing $M^{(\ell^* \rightarrow L)}$ for the summed Jacobian of all attention and MLP contributions downstream of ℓ^* , the first-order logit response to an injected v decomposes as

$$\Delta \ell = \underbrace{\widetilde{W}_U^* v}_{\text{direct}} + \underbrace{\widetilde{W}_U^* M^{(\ell^* \rightarrow L)} v}_{\text{indirect}} + O(\|v\|^2). \quad (4)$$

For $v = v^\perp$ the direct term vanishes on T identically, so any change in the T -restricted logits — and, to the extent that behavior on a deception benchmark is mediated by T , any change in measured behavior — must travel through downstream computation (Theorem 6.1 of the supplementary proof).

3.4 The depth statistic

Let $\beta(v)$ be a benchmark score under intervention v and $\Delta(v) = \beta(v) - \beta(0)$. The depth statistic is

$$\rho(v_{\text{dec}}, T) = \frac{\Delta(v^\perp)}{\Delta(v_{\text{dec}})}. \quad (5)$$

Under a pure vocabulary-suppression account, $\rho \approx 0$ once T exhausts the readout-aligned mass of v_{dec} ; under a pure upstream-concept account, $\rho \approx 1$. We report ρ with paired bootstrap confidence intervals, its stability σ_T across independent constructions of T , and a per-prompt analogue ρ_η defined on the honest-shift

$$\eta(v) = [\mu_{T^+}(v) - \mu_{T^-}(v)] - [\mu_{T^+}(0) - \mu_{T^-}(0)], \quad (6)$$

where $\mu_{T^\pm}$ is the mean logit over honest-coded (T^+) and deceptive-coded (T^-) tokens at the answer position.

Token-set constructions. We instantiate T three ways. *Curated*: a fixed lexicon of 32 honest and 32 deceptive lemmas expanded to single-token surface variants. *Statistical*: the top-32 tokens per side by mean first-position logit shift between honest- and deceptive-prompted contexts on held-out prompts. *Aligned- k* : the top- k tokens by $|\widehat{W}_U^* v_{\text{dec}}|$ — the k tokens the steering vector moves most through the direct path. The aligned construction is the strongest form of the test: it removes exactly the direct-readout content of v_{dec} itself, and sweeping k traces how the surviving effect $\rho(k)$ decays as more of the direct path is excised.

Controls. Three controls separate the unembedding subspace from generic perturbation. (i) *Random projection*: project out a random k -dimensional subspace (three seeds), matching the dimension count of the aligned-64 set. (ii) *Norm matching*: rescale $v^\perp$ to $\|v_{\text{dec}}\|$, separating direction from magnitude. (iii) *Parallel injection*: inject $v^\parallel$ alone, the readout-aligned rank- $\leq k$ component; under the suppression account this component should carry the behavioral effect.

4 Experimental setup

Model. All experiments use Llama-3-8B-Instruct (Grattafiori et al., 2024) ($L = 32$, $d = 4096$, $V = 128,256$, RMSNorm) in bfloat16. Projections and certificates are computed in float64.

Data and splits. TruthfulQA (Lin et al., 2022) provides the behavioral benchmark in its multiple-choice form (817 questions). A seeded shuffle assigns 120 questions to a validation slice used only for operating-point selection, 200 to construct mass-mean statements, and the remaining 497 to the held-out test slice on which every headline number is computed. The CAA vector is built from 256 Alpaca instructions (Taori et al., 2023) rendered under an honest and a deceptive system prompt; a disjoint block of 96 instructions serves as held-out fluency text for the coherence gate, and a further disjoint block of 128 feeds the statistical token set. Vector-construction data and evaluation data share no prompts.

Steering-vector families. We decompose two constructions head to head. The CAA vector (Rimsky et al., 2024) is the difference of mean final-token residuals between honest- and deceptive-prompted renderings, computed per layer. The *mass-mean* vector (Marks and Tegmark, 2024) is

the difference of mean residuals between true and false Q:/A: statements assembled from the 200 reserved TruthfulQA questions, computed at the same candidate layers. Each family is analyzed at its own operating point with its own projections; the unembedding apparatus is shared.

Coherence-gated operating points. For each family we sweep $\ell \in \{12, 14\}$ and $\alpha \in \{0.5, 1, 1.5, 2, 3, 4\}$ and score each setting two ways: the validation MC2 improvement, and the ratio of held-out per-token perplexity under steering to baseline on the 96 reserved Alpaca instructions. A setting is *coherent* if its perplexity ratio is at most 1.5; the operating point is the coherent setting with the largest validation MC2 gain. This gate exists because teacher-forced multiple-choice scores are nearly blind to generation collapse: a sufficiently large α raises MC2 while reducing free generation to gibberish (Section ??), and a sweep that maximizes MC2 alone selects exactly that regime. The gate selects $(\ell^*=12, \alpha^*=3)$ for CAA and $(\ell^*=12, \alpha^*=4)$ for mass-mean.

Scoring. TruthfulQA MC follows the original protocol: the total log-probability of each answer choice under the prompt Q: {question}\nA:, with MC1 the accuracy of the argmax choice and MC2 the normalized probability mass on true choices. The intervention is applied at every token position. At the answer position we additionally record the logits of every token in the union of all token sets, from which the honest-shift η and the spillover readouts are computed without further forward passes.

Conditions. For each family, the headline matrix evaluates on the 497 test questions: a shared unsteered baseline; the family’s vector v_{dec} ; $v^\perp$ for aligned- k with $k \in \{16, 64, 256, 1024\}$; $v^\perp$ for the curated and statistical sets; the parallel component $v^\parallel$ (aligned-64); the norm-matched $v^\perp$ (aligned-64); and random 64-dimensional projections (three seeds). All conditions within a family share that family’s (ℓ^*, α^*) .

Out-of-distribution probe. The first 150 test questions are paraphrased by the unsteered model (greedy decoding, meaning-preserving instruction) and rescored under baseline, v_{dec} , and $v^\perp$ (aligned-64) per family, giving an OOD depth ratio ρ_{OOD} . The spillover readout evaluates η on stem-disjoint synonym sets that were never projected out. A logit-lens trajectory (nostalgibraist, 2020) tracks the curated- T readout across layers under v_{dec} , $v^\perp$, and $v^\parallel$ to localize where the honest-shift re-emerges.

Reproducibility. The pipeline is staged and checkpointed with fixed seeds; all code, token lists, certificates, and the formal proof are in the repository. The zero-direct-effect certificates $\max_t |(Av^\perp)_t|$ are at most 7×10^{-15} logits across every family and token set — machine precision in float64 — against pre-projection direct effects of up to 0.77 logits per unit α .

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